

Supplementary Tables

Preregistration: Do Commercial LLM APIs Degrade Answer Quality Under Load? A 21-Day Within-Provider Longitudinal Measurement of Eight Models.

This file lists the files registered alongside the form, then gives seven tables that the form's text fields cannot hold in tabular form. Each table caption names the field it belongs to.

Registered files

Five files are registered together with the form, this one included. They are frozen at submission and cannot be edited afterwards. The same material also lives in a public repository, which remains editable; where the two ever disagree, the registered files are the authoritative version.

prereg_supplementary_tables.pdf (this file)

Seven tables the form's text fields cannot hold in tabular form. Each caption names the form field the table belongs to.

prompts.py

Prompt construction and the answer parser, fixed before execution. Registering it is the commitment made under blinding: scoring is condition-blind because `parse_letter()` reads only the response string and does not call an LLM. `DIRECT_SYSTEM` is the quality-probe system prompt; `make_nonce()` is the cache-blocking token described under Study design.

item_bank_ids.csv

The 300 items of the fixed bank, by MMLU-Pro identifier and subject, 50 per subject. Registering the list is the commitment made under blinding: the bank was finalized before any peak data were collected. Item text and answer keys are not reproduced; MMLU-Pro is distributed by TIGER-Lab under the MIT license and the identifiers suffice to reconstruct the bank. The `bank_order` column is each item's position in the fixed bank. The three presentation groups described under Study design are consecutive blocks of 100 in this order, so that column determines the counterbalancing together with the rotation rule stated there.

12_condition_power.py

The script that computes the required item counts and the coverage reported under Sample size rationale.

condition_power.md

The output of that script for the registered design: 21 completed days, $k = 3$, a 300-item bank, power 0.80, two-sided alpha 0.05. Tables 3 and 4 below are drawn from it.

Tables

Table 1. Models measured

Design Plan / Study design

Provider	Model	Region	Reasoning setting
OpenAI	gpt-5.6-luna	US	Disabled via effort=none
Google	gemini-3.5-flash-lite	US	Cannot be fully disabled; minimal is the floor
Anthropic	claude-haiku-4-5-20251001	US	Off by default
DeepSeek	deepseek-v4-flash	China	Cannot be disabled; run enabled
Qwen	qwen3.7-flash-2026-07-15	China	Run enabled
Naver	HCX-DASH-002	Korea	Not applicable

Primary set. The tier-comparison set adds gpt-5.6-sol (OpenAI) and claude-sonnet-5 (Anthropic).

Table 2. Condition definitions, taken from the providers' published windows*Design Plan / Study design*

Arm	Peak definition	Source	Peak slots (UTC)
US	Weekdays 08:00-14:00 ET	Published by Anthropic	12, 15
China	Daily 01:00-04:00, 06:00-10:00 UTC	Published by DeepSeek	03, 06, 09
Korea	Daily 09:00-12:00, 14:00-18:00 KST	China definition, time-zone adjusted	00, 06

Every slot outside a peak window is off-peak; no intermediate category is defined. Because the US definition covers weekdays only, weekend slots at the same hours count as off-peak.

Table 3. Required item counts and coverage by arm*Sampling Plan / Sample size rationale*

Arm	Peak trials	Off-peak trials	Accuracy	Self-consistency	Reasoning tokens	Coverage
US	10.0	46.0	282	206	n/a	0.157
China	21.0	35.0	250	200	22	0.168
Korea	14.0	42.0	263	202	n/a	0.164

$k = 3$, 21 completed days, 300-item bank, power 0.80, two-sided alpha 0.05. The accuracy, self-consistency and reasoning-token columns give the item count required at $\tau = 0.15$. Reasoning-token cells are empty for the US and Korea because those models do not report reasoning tokens in usage. Accuracy is the binding constraint in all three arms. Computed by analysis/12_condition_power.py.

Table 4. Coverage by repetition count k *Sampling Plan / Sample size rationale*

k	US	China	Korea	Minimum	Increment
2	0.141	0.159	0.152	0.141	
3	0.157	0.168	0.164	0.157	+0.016
4	0.165	0.173	0.169	0.165	+0.008
5	0.169	0.175	0.173	0.169	+0.004

Bank size and schedule are fixed, so each k admits one attainable coverage. $k = 3$ is the point at which the gain flattens; call count rises in proportion to k . The declared coverage, τ at or below 0.15, is the $k = 3$ minimum rounded down.

Table 5. Primary outcome variables*Variables / Measured variables*

Variable	Definition	Models measured
Self-consistency	Share held by the modal answer when the same item is answered three times within one trial	All 8 models
Reasoning tokens	Reasoning-token count reported in the response usage	deepseek-v4-flash, qwen3.7-flash
Accuracy	Proportion of parsed answers matching the correct answer	Measured on all 8 models; interpreted only for models with headroom

Table 6. Mixed-effects model specifications (secondary analysis)*Analysis Plan / Statistical models*

Metric	Unit	Model
Self-consistency	Trial	Linear mixed model; condition as fixed effect, item and date as random intercepts
Reasoning tokens	Natural log of the trial mean	Same specification
Accuracy	Call	Logistic mixed model; same fixed and random effects

On non-convergence or a singular fit, the date random effect is dropped first; if it still fails, only the item random intercept is retained. This order is fixed in advance.

Table 7. Holm correction families, by model*Analysis Plan / Inference criteria*

Model	Metrics in the family	Number of tests
gpt-5.6-luna, gemini-3.5-flash-lite, claude-haiku-4-5	Self-consistency, accuracy	2
deepseek-v4-flash, qwen3.7-flash	Self-consistency, accuracy, reasoning tokens	3
HCX-DASH-002	Self-consistency	1
gpt-5.6-sol, claude-sonnet-5	Self-consistency	1

Correction is applied within each model, never pooled across models, because the study consists of eight independent longitudinal measurements. Accuracy is excluded for HCX-DASH-002 (0.18 on the bank, near the floor) and for the two comparison-set models (0.82, near the ceiling).